Matrices: Inverses & Applications

Questions 1–64

Question 1. It is given that A and B are non-singular square matrices. Given also that $AB = I$, write down how A and B are related.

Question 2. It is given that A is a square matrix. Given also that A is singular, explain why A does not have an inverse matrix.

Question 3. In each case, find the inverse of the 2D matrix:

a) $\begin{pmatrix} 3 & 7 \\ 2 & 5 \end{pmatrix}$

b) $\begin{pmatrix} 8 & 13 \\ -3 & -5 \end{pmatrix}$

c) $\begin{pmatrix} 9 & 1 \\ 28 & 3 \end{pmatrix}$

d) $\begin{pmatrix} 6 & 17 \\ 2 & 6 \end{pmatrix}$

e) $\begin{pmatrix} -8 & -7 \\ 11 & 10 \end{pmatrix}$

f) $\begin{pmatrix} 12 & 20 \\ 5 & 10 \end{pmatrix}$

g) $\begin{pmatrix} -2 & 9 \\ 1 & 3 \end{pmatrix}$

h) $\begin{pmatrix} 6 & -5 \\ -25 & 0 \end{pmatrix}$

i) $\begin{pmatrix} -3 & 8 \\ -9 & 7 \end{pmatrix}$

j) $\begin{pmatrix} 20 & -30 \\ -50 & -40 \end{pmatrix}$

Question 4. It is given that $A = \begin{pmatrix} k & 4 \\ 2 & 3 \end{pmatrix}$, where k is a constant.

a) Find the value of k such that A is singular.

b) Given that A is not singular, find A^{-1} in terms of the constant k.

Question 5. It is given that $A = \begin{pmatrix} k & 2 \\ 2 & k \end{pmatrix}$, where k is a positive constant.

a) Find the value of k such that A is singular.

b) Given that A is not singular, find A^{-1} in terms of the constant k.

Question 6. It is given that $A = \begin{pmatrix} k+1 & -2\sqrt{2} \\ -2\sqrt{2} & k-1 \end{pmatrix}$, where k is a negative constant.

a) Find the value of k such that A is singular.

b) Given that A is not singular, find A^{-1} in terms of the constant k.

Question 7. Each of the following 2D matrices represents a linear transformation. In each case, i) fully describe the transformation represented by the matrix, ii) write down its inverse matrix, and iii) fully describe the transformation represented by its inverse matrix.

a) $\begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$

b) $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

c) $\begin{pmatrix} 8 & 0 \\ 0 & 1 \end{pmatrix}$

d) $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

e) $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

f) $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$

g) $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

h) $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$

i) $\begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$

j) $\begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix}$

k) $\begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$

l) $\begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$

Question 8. Each of the following 2D matrices represents a shear. In each case, i) fully describe the transformation represented by the matrix, ii) write down its inverse matrix, and iii) fully describe the transformation represented by its inverse matrix.

a) $\begin{pmatrix} 1 & 4 \\ 0 & 1 \end{pmatrix}$

b) $\begin{pmatrix} 1 & 0 \\ 6 & 1 \end{pmatrix}$

c) $\begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}$

d) $\begin{pmatrix} 1 & 0 \\ -\frac{2}{5} & 1 \end{pmatrix}$

Question 9. It is given that A is the 2D matrix that represents a rotation, anticlockwise about the origin by angle θ. It is also given that B is the 2D matrix that represents a rotation, clockwise about the origin by angle θ.

a) Write down the matrix A.
b) Write down the matrix B.
c) Use $AB = I$ to derive the trigonometric identity $\sin^2\theta + \cos^2\theta = 1$.

Question 10. It is given that $A = \begin{pmatrix} 2 & 3 \\ 3 & 5 \end{pmatrix}$.

a) Find A^{-1}.
b) The point P is mapped to the point $Q(1, 3)$ under the transformation represented by the matrix A. Find the coordinates of P.

Question 11. It is given that $A = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$.

a) Find A^{-1}.

b) The point P is mapped to the point $Q(-3, 1)$ under the transformation represented by the matrix A. Find the coordinates of P.

Question 12. It is given that $A = \begin{pmatrix} 3 & 2 \\ 1 & 1 \end{pmatrix}$ and $C = \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}$. Find the matrix B such that $AB = C$.

Question 13. It is given that $B = \begin{pmatrix} 4 & 3 \\ 6 & 5 \end{pmatrix}$ and $C = \begin{pmatrix} 6 & 2 \\ 4 & 1 \end{pmatrix}$. Find the matrix A such that $AB = C$.

Question 14. It is given that $A = \begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix}$ and $AB = \begin{pmatrix} 2 & -1 & 3 \\ -2 & -4 & 5 \end{pmatrix}$. Find the matrix B.

Question 15. It is given that $B = \begin{pmatrix} 9 & -1 \\ -2 & 3 \end{pmatrix}$ and $AB = \begin{pmatrix} 34 & -1 \\ -24 & 11 \\ 16 & 1 \end{pmatrix}$. Find the matrix A.

Question 16. It is given that $B = \begin{pmatrix} 5 & 3 \\ 2 & 6 \end{pmatrix}$ and $AB = \begin{pmatrix} 9 & 15 \\ 33 & 39 \\ 11 & 21 \\ 26 & 54 \end{pmatrix}$. Find the matrix A.

Question 17. It is given that $A = \begin{pmatrix} 2 & 4 \\ -9 & 8 \end{pmatrix}$.

a) Find A^{-1}.

b) Given that $AB = \begin{pmatrix} 2 & -2 \\ -35 & 61 \end{pmatrix}$ and that $C = 3A - 4B$, find the matrix C.

Question 18. It is given that $A = \begin{pmatrix} 2 & 3 \\ -1 & -2 \end{pmatrix}$. Show that the matrix A is self-inverse.

Question 19. The triangle T is mapped to the triangle T' by the matrix $A = \begin{pmatrix} 2 & 1 \\ 13 & 7 \end{pmatrix}$. The triangle T' has vertices at $(2, 3)$, $(4, -5)$ and $(-1, -1)$. Find the coordinates of the vertices of T.

Question 20. Given $A = \begin{pmatrix} p & 2q \\ 4q & p \end{pmatrix}$ and $C = \begin{pmatrix} 3p & q \\ 5q & 2p \end{pmatrix}$, find the matrix B such that $AB = C$.

Question 21. Given $A = \begin{pmatrix} 3p & 4q \\ 2q & -p \end{pmatrix}$ and $C = \begin{pmatrix} 5p & -2q \\ -4q & p \end{pmatrix}$, find matrix B such that $AB = C$.

Question 22. It is given that $\begin{pmatrix} 3k & -5 \\ -k & 7 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 6k \\ -3 \end{pmatrix}$. Write x and y in terms of the constant k.

Question 23. It is given that $A = \begin{pmatrix} 3 & k \\ -2 & 4 \end{pmatrix}$, where k is a constant. Find the value of k such that $2A^{-1} + A = 7I$.

Question 24. It is given that $A = \begin{pmatrix} 5 & 7 \\ 2 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 4 & 3 \\ 7 & 6 \end{pmatrix}$.

a) Show that A and B are both non-singular.

b) Show that $(AB)^{-1} = B^{-1}A^{-1}$.

Question 25. It is given that $A = \begin{pmatrix} 8 & 2 \\ 7 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} -1 & -1 \\ 3 & 2 \end{pmatrix}$.

a) Find A^{-1}.
b) Find B^{-1}.
c) Hence find $(AB)^{-1}$.

Question 26. It is given that $A = \begin{pmatrix} 3 & 2 \\ 2 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 5 & 3 \\ 3 & 2 \end{pmatrix}$ and $C = \begin{pmatrix} 6 & 7 \\ 3 & 3 \end{pmatrix}$.

a) Find A^{-1}.
b) Find B^{-1}.
c) Find C^{-1}.
d) Hence find $(ABC)^{-1}$.

Question 27. In each case, find the value(s) of the constant k such that the matrix is self-inverse:

a) $\begin{pmatrix} k & -4 \\ 2 & 3 \end{pmatrix}$

b) $\begin{pmatrix} k & -35 \\ 1 & -k \end{pmatrix}$

c) $\begin{pmatrix} -3 & k \\ -2k & 3 \end{pmatrix}$

d) $\begin{pmatrix} k+1 & 4 \\ -k & -3 \end{pmatrix}$

e) $\begin{pmatrix} -\frac{\sqrt{2}}{2} & 2k-1 \\ -4k & \frac{\sqrt{2}}{2} \end{pmatrix}$

Question 28. It is given that A is a self-inverse matrix. Prove that $A^2 = I$.

Question 29. It is given that $A = \begin{pmatrix} a & 6 \\ b & 2 \end{pmatrix}$ and $B = \begin{pmatrix} -3 & 2 \\ -4 & 6 \end{pmatrix}$ and $AB = kI$, where a, b, c and k are constants.

a) Find the values of a, b, c and k.

b) Hence determine A^{-1}.

Question 30. It is given that A and B are 2D matrices such that $A^{-1}B^{-1} = \begin{pmatrix} \frac{5}{6} & -\frac{1}{3} \\ -\frac{2}{3} & \frac{2}{3} \end{pmatrix}$.

a) Find BA.

b) Given also that $B = \begin{pmatrix} 1 & 0 \\ 1 & \frac{3}{2} \end{pmatrix}$, find A.

Question 31. It is given that A and B are square matrices such that $\det A = 5$ and $\det B = \frac{2}{3}$. Find:

a) $\det(A^{-1})$

b) $\det(B^{-1})$

c) $\det((AB)^{-1})$

Question 32. It is given that A and B are square matrices such that $\det(A^{-1}) = 4$ and $\det B = -2$. Find:

a) $\det(A^4)$

b) $\det(AB)$

Question 33. It is given that A and B are square matrices such that $\det(A^{-1}) = -\frac{1}{5}$ and $\det(AB) = 40$. Find $\det(B^5)$.

Question 34. It is given that $A = \begin{pmatrix} 10 & -6 & 2 \\ 3 & -2 & 1 \\ -4 & 4 & -2 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 2 & 1 \\ -1 & 6 & 2 \\ -2 & 8 & 1 \end{pmatrix}$.

a) Find AB.

b) Hence find A^{-1}.

Question 35. It is given that $A = \begin{pmatrix} p & 0 & q \\ 2 & 1 & q \\ -2 & 0 & r \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 0 & -2 \\ 4 & 4 & \frac{1}{3} \\ -1 & 0 & 4 \end{pmatrix}$ and $AB = kI$, where p, q and r are constants.

a) Find the values of p, q and r.

b) Hence find A^{-1}.

Question 36. In each case, find the inverse of the 3D matrix:

a) $\begin{pmatrix} 2 & 1 & 0 \\ 7 & 1 & 2 \\ -3 & -3 & 1 \end{pmatrix}$

b) $\begin{pmatrix} 2 & 1 & -2 \\ 0 & 0 & 1 \\ 4 & 3 & -1 \end{pmatrix}$

c) $\begin{pmatrix} 4 & 0 & 3 \\ 4 & 3 & 6 \\ 5 & -1 & 3 \end{pmatrix}$

Question 37. It is given that $A = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 1 & -3 \\ 1 & 2 & -3 \end{pmatrix}$.

a) Find A^{-1}.

b) The point P is mapped to the point $Q(1, 3, -2)$ under the transformation represented by the matrix A. Find the coordinates of P.

Question 38. It is given that $A = \begin{pmatrix} -\frac{3}{2} & 5 & 0 \\ 1 & 3 & -2 \\ -1 & 0 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 6 & -10 & -10 \\ 0 & -3 & -3 \\ 3 & -5 & -\frac{19}{2} \end{pmatrix}$. Show that $A = kB^{-1}$, where k is a constant to be found.

Question 39. In each case, find the inverse of the 3D matrix in terms of the constant k:

a) $\begin{pmatrix} k & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix}$

b) $\begin{pmatrix} 2 & 0 & k \\ 3 & 1 & 0 \\ 2 & 1 & 4 \end{pmatrix}$

c) $\begin{pmatrix} 2k & -1 & 3 \\ k & -5 & 1 \\ k & -2 & 4 \end{pmatrix}$

d) $\begin{pmatrix} -1 & 2 & -1 \\ k & 3 & k \\ -1 & 2 & 1 \end{pmatrix}$

Question 40. It is given that $A = \begin{pmatrix} 1 & -1 & k \\ k & 9 & 18 \\ 3 & 0 & 27 \end{pmatrix}$.

a) Show that for all values of k, A is a non-singular matrix.

b) Find A^{-1}.

Question 41. It is given that $A = \begin{pmatrix} 5-k & 4 & -2 \\ k & -2 & 5 \\ -1 & k+1 & -5 \end{pmatrix}$. It is also given that A is a non-singular matrix. Find A^{-1} and state any restrictions on the value of k.

Question 42. It is given that $A = \begin{pmatrix} -2-k & -3 & 1 \\ -1 & -3 & k+2 \\ -1 & -2 & k+3 \end{pmatrix}$. It is also given that A is a non-singular matrix. Find A^{-1} and state any restrictions on the value of k.

Question 43. It is given that $A = \begin{pmatrix} 1 & 8 & k \\ 0 & -3 & -2 \\ 0 & k & 3 \end{pmatrix}$, where k is a constant. It is also given that A is self-inverse. Find the value of k.

Question 44. Use a matrix method to solve the following systems of linear equations:

a) $2x + 3y = 5$, $x + y = 4$
b) $3x + y = -3$, $x + 2y = 9$
c) $4x - y = 15$, $8x - 3y = 1$
d) $2y - 6x = -7$, $9x - 4y = -10$

Question 45. Use a matrix method to find the two values of k for which the equations $2x+9ky=6$ and $kx+18y=6$ do not have a unique solution. Interpret the geometrical significance for each of your values of k.

Question 46. Use a matrix method to find the two values of k for which the equations $8kx+48y=5$ and $3x+2ky=3$ do not have a unique solution. Interpret the geometrical significance for each of your values of k.

Question 47. It is given that $A = \begin{pmatrix} 0.2 & 0 & -0.3 \\ 0 & -0.2 & 0.3 \\ 0.1 & -0.2 & -0.1 \end{pmatrix}$ and $A^{-1} = \begin{pmatrix} 8 & 6 & -6 \\ 3 & 1 & -6 \\ 2 & 4 & -4 \end{pmatrix}$. It is also given that $2x - 3z = 40$, $-2y + 3z = -60$ and $x - 2y - z = 90$.

a) Without solving the equations, explain how you know that they have a unique solution.

b) Find the unique solution of the three simultaneous equations.

Question 48. Use a matrix method to solve the following systems of linear equations:

a) $x + y + 2z = 1$, $2x + y + z = 3$, $3x - y + 4z = 2$

b) $x + y = 5$, $x - y + z = 8$, $2x + 5z = 3$

c) $y - 3z = -7$, $4z - x = 3$, $z - y - x = -2$

Question 49. Use a matrix method to solve the following system of linear equations: $2a + b + 2c - 3d = 8$, $-a + 4b + 3c + 5d = 14$, $6a - 2b + c - 8d = 11$, $-9a - 7b + 3c + 2d = 6$.

Question 50. It is given that $A = \begin{pmatrix} k & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & k \end{pmatrix}$.

a) Find A^{-1} in terms of k.

b) Use your answer from a) to solve $3x + y + z = -6$, $x + z = 18$, $x + 3z = -12$.

Question 51. It is given that $A = \begin{pmatrix} -5 & 2 & 3 \\ 1 & 3 & 2 \\ k & 2 & 3 \end{pmatrix}$.

a) Find A^{-1} in terms of k.

b) Use your answer from a) to solve $-5x + 2y + 3z = 2$, $x + 3y + 2z = -5$, $5x + 2y + 3z = 9$.

Question 52. It is given that A, B and C are non-singular square matrices. Simplify each of the following expressions:

a) $C^{-1}CAB^{-1}B$

b) $B(AB)^{-1}$

c) $(CA^{-1})^{-1}C$

d) $C(ABC)^{-1}A$

e) $(B^{-1}A)^{-1}(C^{-1}B)^{-1}$

f) $(AB^{-1}C)^{-1}(C^{-1}BA^{-1})^{-1}$

g) $(ACBA^{-1})^{-1}(AB^{-1}C^{-1}A^{-1})^{-1}$

Question 53. It is given that A, B and C are non-singular square matrices. Given that $AB = C$, prove that $B = A^{-1}C$.

Question 54. It is given that A, B and C are non-singular square matrices. Given that $ABC = A$, prove that $B = C^{-1}$.

Question 55. Isla believes that for two square matrices, A and B, the inverse of $B^{-1}A$ is BA^{-1}. Her proof is shown below:

$$(B^{-1}A)(BA^{-1}) = B^{-1}ABA^{-1} = B^{-1}BAA^{-1} = I = I$$

a) Identify the error in Isla's working.
b) Write down the correct inverse of $B^{-1}A$.
c) Write down a correct proof of your answer to b).

Question 56. It is given that A, B and C are non-singular square matrices. Given that $B = A^{-1}CA$, prove that $C = ABA^{-1}$.

Question 57. It is given that A and B are non-singular square matrices. Given that $ABA = I$, prove that $B = A^{-1}A^{-1}$.

Question 58. It is given that A and B are non-singular square matrices. Prove that $(AB)^{-1} = B^{-1}A^{-1}$.

Question 59. It is given that A, B and C are non-singular square matrices. Prove that $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$.

Question 60. It is given that A, B and C are non-singular square matrices. Prove that $(A^{-1}B^{-1}C^{-1})^{-1} = CBA$.

Question 61. It is given that A and B are non-singular square matrices. Prove that $\det(A^{-1}) = \frac{1}{\det A}$.

Question 62. It is given that A is a non-singular square matrix. Given that $(AB)^T = B^T A^T$, show that $(A^T)^{-1} = (A^{-1})^T$.

Question 63. It is given that $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $B = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.

a) Show that $AB = I$.
b) Show that $BA = I$.

Question 64. It is given that $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Derive that $A^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.